Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels

Milestone 1: Project Orientation & Data Loading

Infosys Springboard Virtual Internship 6.0

Progress Report – Milestone 1

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1. Project Orientation

The project **Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels** focuses on analyzing historical hotel booking data to understand revenue performance and booking behavior in the hospitality industry. Hotels face challenges such as fluctuating demand, seasonal variations, cancellations, and pricing decisions, all of which directly impact revenue.

This project applies data analytics and business intelligence techniques using Power BI to convert raw booking data into meaningful insights. The analysis is designed to support hotel management in making informed, data-driven decisions for revenue optimization and operational planning.

2. Abstract

This project aims to analyze hotel booking data to identify revenue trends, demand patterns, and customer behavior using business intelligence tools. The dataset is loaded, reviewed, and prepared to ensure accuracy and consistency. Through structured data analysis, DAX-based calculations, and visualization techniques, the project seeks to deliver actionable insights that support effective hotel revenue management. The project also establishes a strong foundation for AI-driven analysis in later milestones.

3. Objectives

- The primary objectives of this project are:
- To analyze historical hotel booking data for revenue-related insights
- To study booking patterns and cancellation behavior
- To identify seasonal trends and demand fluctuations
- To support data-driven decision-making in hotel revenue management
- To prepare a strong analytical foundation for DAX, visualization, and AI-assisted insights in future milestones

4. Dataset Description

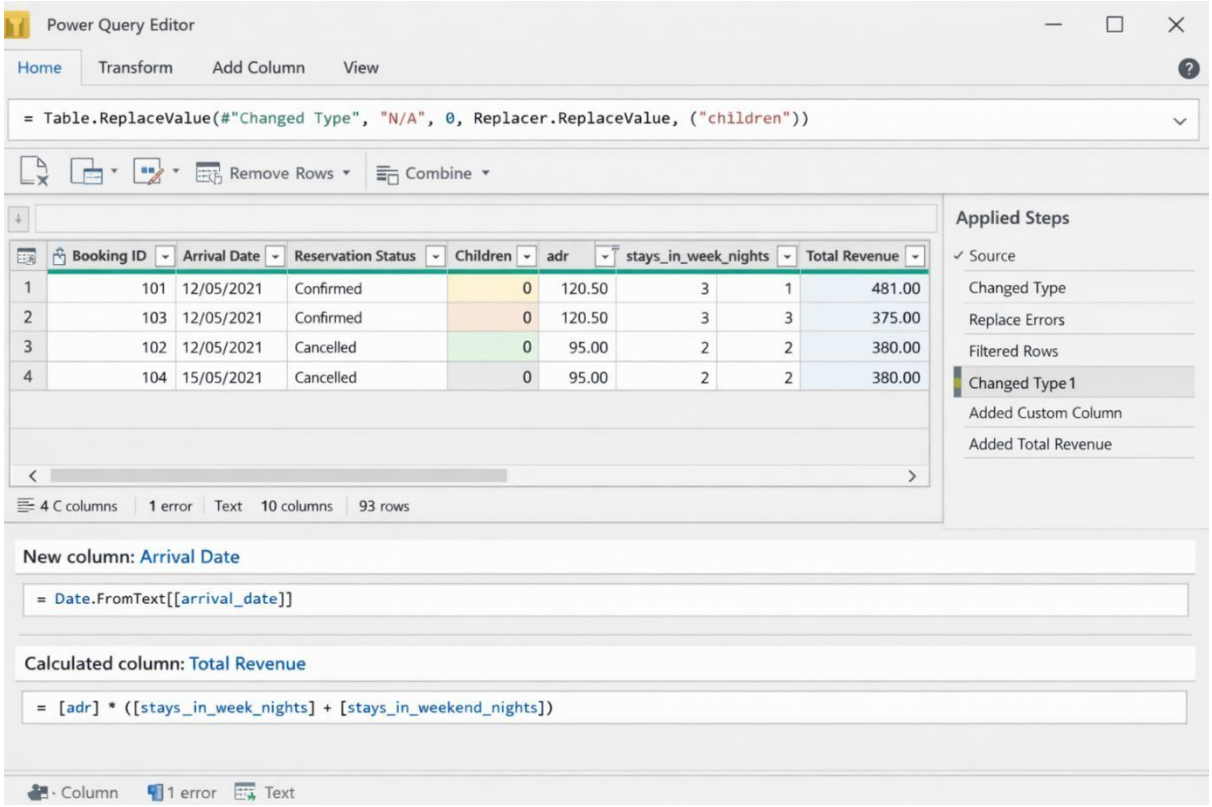
The dataset used in this project is a cleaned hotel booking dataset containing historical reservation information. It includes attributes such as hotel type, arrival dates, number of guests, length of stay, average daily rate (ADR), and reservation status.

This dataset provides a comprehensive view of hotel operations and customer behavior, making it suitable for analyzing revenue performance, booking trends, and demand patterns.

5. Data Cleaning

As part of Milestone 1, the dataset was reviewed and cleaned using Power Query Editor to ensure data accuracy and reliability. Data cleaning involved handling invalid or missing values, correcting data types, and ensuring uniform formatting across important fields such as dates, pricing, and guest information.

Additionally, a proper arrival date column was created, and total revenue was calculated based on stay duration and average daily rate. These steps ensured that the dataset is structured, consistent, and ready for data modeling, DAX calculations, and visualization in subsequent milestones.



6. Individual Work

This documentation has been prepared individually as part of Milestone 1. It reflects personal understanding of the project scope, dataset, and analytical approach. The individual preparation of this document demonstrates independent learning and responsibility in completing the assigned milestone.

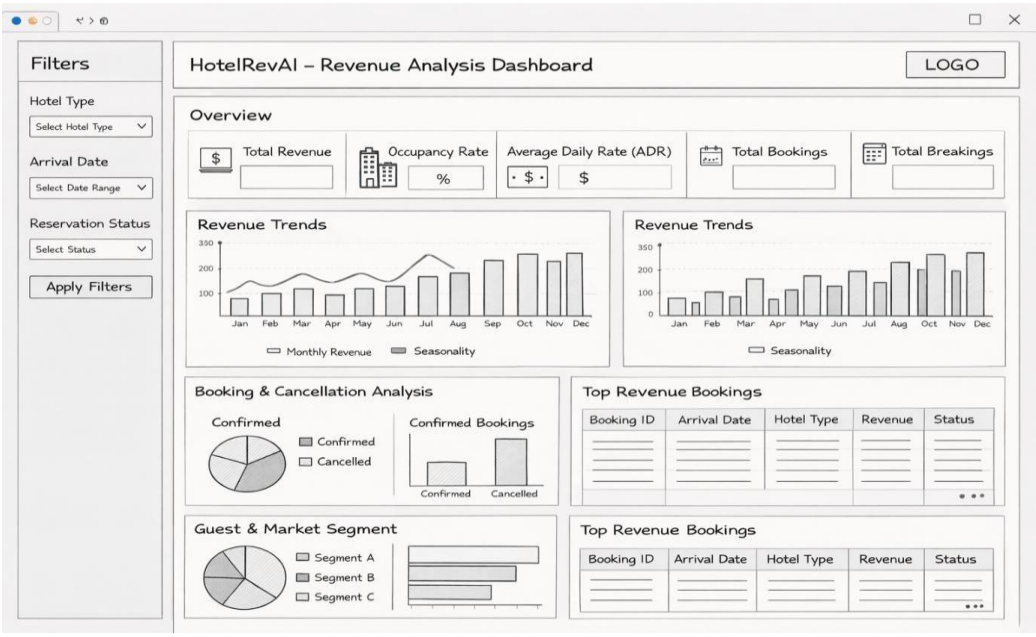
7. Next Steps

The next phase of the project will focus on:

- Designing an optimized data model in Power BI
- Creating DAX measures for key business metrics such as revenue and booking trends
- Developing interactive dashboards for visualization and storytelling
- Exploring AI-driven insights using Power BI features to enhance decision-making

8. Proposed Dashboard Design (Mockup)

The following dashboard mockup illustrates the proposed layout and key KPIs for the HotelRevAI project. It is a conceptual design created for planning purposes, and the final Power BI dashboard will be developed and refined in later milestones.



9. Project Enhancement Tips (Future Scope)

To further enhance the **Infosys_HotelRevAI** project, the following improvements can be implemented in later stages:

- Development of advanced DAX measures such as occupancy rate, revenue growth, and cancellation impact analysis
- Detailed seasonality analysis to identify peak and off-season trends
- Use of AI visuals in Power BI, such as Key Influencers and Decomposition Tree, to uncover revenue drivers
- Revenue forecasting to support future demand planning
- Storytelling dashboards designed for hotel managers and decision-makers

These enhancements will increase the analytical depth, business value, and practical relevance of the project.

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